

Signals and Systems

Mumbai University Engineering Syllabus - Pre-loaded Reference for Conscious Commute.

Module 1: Introduction to Signals and Systems

Continuous-time and discrete-time signals, operations, classification of signals (even/odd, periodic, energy/power). Systems: properties, linearity, causality, time-invariance, stability.

Module 2: Linear Time-Invariant (LTI) Systems

Continuous-time and discrete-time LTI systems, convolution integral and convolution sum, properties of LTI systems.

Module 3: Fourier Analysis of Continuous-Time Signals

Continuous-Time Fourier Series (CTFS), Fourier Transform (CTFT), properties, frequency response of LTI systems.

Module 4: Fourier Analysis of Discrete-Time Signals

Discrete-Time Fourier Transform (DTFT), properties, system analysis using DTFT.

Module 5: Laplace Transform in Signal Analysis

Unilateral and Bilateral Laplace transforms, region of convergence (ROC), system function, stability analysis.

Module 6: Z-Transform

Z-transform, ROC, properties, inverse Z-transform, analysis of discrete-time LTI systems.